



RAMCO INSTITUTE OF TECHNOLOGY

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Department of Computer Science and Engineering

Academic Year 2022 – 2023 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. CSE

Course Code & Title: CS3452 Theory of Computation

Name of the Faculty member (s): Mrs.S.Manjula

Innovative Practice Description

Unit / Topic: Unit V / Tractable and Intractable problems

Course Outcome: CO 5

Topic Learning Outcome: TLO 18

Activity Chosen: Learning by Teaching

Justification:

The main objective of the learning by teaching activity was to promote active learning and knowledge consolidation among the students. By assigning topics to the students and having them teach those topics to their peers, the activity aimed to reinforce their understanding, improve communication skills, and foster a collaborative learning environment.

- **Time Allotted for the Activity:** 35 minutes

Details of the Implementation:

- Prior to the session, the instructor asked the students about willingness about teaching to the particular topic.
- Sundara Mahalingam P shown interest to take class.
- Student was assigned a Tractable and Intractable problems topic based on their relevance and complexity.
- Student was given time to prepare their assigned topics. He was encouraged to gather information from various sources, organize the content effectively, and plan their teaching approach.
- During the session, students took turns presenting their assigned topics to their classmates.
- He utilized different teaching techniques such as presentations, solving problem, and interactive activities to engage the class mates.
- After that, a question-and-answer session was held to encourage active participation and clarify any doubts.
- This allowed students to engage in discussions and exchange additional insights related to the topic.

CO – PO / PSO mapping:

CO	PO1	PO2	PO4	PO9	PO10	PSO1
CO 5	2	2	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO4	PO9	PO10	PSO1
	2	2	1	1	1	1
Justification for correlation	Apply mathematical knowledge to find computational solutions to various problems.	To analyze complex engineering problems using Tractable and Intractable problems	Analyze the problem to find the solution	To work as an individual	To Communicate effectively on complex engineering activities	Will be able to solve complex computation task

• **Images / Screenshot of the practice:**



Figure 1: Problem solution explanation by Sundara Mahalingam P



Figure 2: Discussion about Tractable and Intractable problems by Sundara Mahalingam P

- **Reflective Critique:**

- ❖ **Feedback of practice from students and other stakeholders:**

- Sundara Mahalingam P informed the instructor that the activity encouraged him to have a deeper understanding of the subject matter and improve his retention of the information.
 - All the students were eagerly listened the class. Students felt they are able to understand the topic.
 - Students highlighted that this practice helped them improve their communication skills, such as clarity, organization, and the ability to present complex ideas in a simplified manner.

- ❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- This activity helped the student to improves communication skills.
 - Students spent their time in self-learning.
 - This activity encouraged the students to share their knowledge with others.

- ❖ **Challenges faced in implementation:**

- Effectively motivate students who are not participating in the activity due to shyness, fear of public speaking, or lack of confidence. Motivate the students to overcome their shyness and engage in active learning.

References:

1. <https://effectiviology.com/protege-effect-learn-by-teaching/>
2. https://www.researchgate.net/publication/351905566_Impact_of_Seminars_on_Student_Soft_Skills_Development
3. https://www.ritrjpm.ac.in/images/computer-science/2022-2023/5_VA_CS3362_Learning%20by%20teaching.pdf